Sam Armstrong

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PERSONAL PROFILE

I am currently finishing my Master's degree in Computer Science, after 3+ years of industry experience building and deploying software solutions that are maintainable, scalable, and genuinely useful. I like problem solving and proactively tackling challenges, and am driven by developing, shipping and iterating rapidly. I would love to work in a workplace that also values these philosophies.

EDUCATION

2025–2026

University of Warwick

MSc Computer Science

Estimated Grade: *Distinction*

Focusing on advanced topics in agentic artificial intelligence, high performance computing, computer security, networks, and computer science theory. I'm awaiting my final results, but all modules have been completed with an average grade of 79%, and my thesis is currently being marked.

2018–2021

University of Sheffield

BSc Artificial Intelligence & Computer Science

Grade: *First Class Honours*

Studied for a Bachelor's degree in AI and Computer Science, achieving a final year grade average of 79.5%. The course focused on the theory and applications of deep learning, software engineering, and system design.

MISCELLANEOUS

- Anthropic London Hackathon *(2023)*
  Created a GitHub bot to automatically add docstrings to functions using the Claude API. DocPilot
- Warwick Hackathon: Whack *(2025)*
  Created an AI learning tool that lectures over a pdf and asks questions.
- Research Presentations *(2021 & 2022)*
  MABS 2022 & Sheffield SIAM-IMA 2021.
- Publication in MABS 2022
- My Sheffield graduate profile

NOTABLE PROJECTS

CozyCasts

Created a platform for streaming AI-generated podcasts. The backend is designed for scalability, while maintaining low latency and cost-conscience, and uses an API/worker pattern to decouple the compute-heavy TTS from the API. The Claude API is used to generate the script, and the TTS uses an open-source model from the Hugging Face Hub.

UFC Almanac

A collection of UFC datasets scraped from the web and regularly updated via CI/CD pipelines, with data transformation pipelines to create structured data for ML training, and deep learning models for predicting fights.

PDF Word Count

Created an extension for Visual Studio Code and Open VSX compatible editors (such as Cursor), which displays the word count of PDF documents in the status bar, with more detailed statistics in the status bar tooltip.

Open VSX Registry VS Code Marketplace GitHub

ENTREPRENEURSHIP

2024–2025

Co-Founder — Ivy

Founded Ivy with two former colleagues, as a spin-off from our previous company (Unify), with the blessing of their CEO. We adapted the existing framework to allow any ML models or code to be converted between frameworks such as PyTorch and JAX. The design of the transpiler involved advanced Python programming, including AST manipulation and function tracing. To create the customer-facing interface, I did significant backend development and DevOps using cloud infrastructure (GCP).

After interviewing with Y Combinator and talking to potential customers, we failed to prove commercial viability, so we ended Ivy as a venture and open-sourced the project - which is available on *GitHub* and *PyPI*.

EMPLOYMENT

2023–2024

Machine Learning Software Engineer — Unify AI (YC W23)

- Developed the multi-backend framework Ivy; creating backend functions and tests, forming the CI, and reviewing PRs from contributors.
- Led an agile team of five, developing a computational graph tracer that allowed any Ivy model to be compiled into an efficient representation.
- Member of the leadership team; attending events to represent the company, and interviewing candidates for part- and full-time roles.

2022–2023

Machine Learning Engineer — Intradiem

- Orchestrated the creation of Java data pipelines for transforming raw data from an AWS S3 datalake into bucketed time-series data for ML applications in both training and inference.
- Led development of a Python micro-service using Flask/MariaDB, containerised with Docker, to document customer/product relationships and make them accessible to services across the business.